Examiner only

6. The main stages in the manufacture of sulfuric acid are given below.

Stage 1 $S + O_2 \longrightarrow SO_2$

Stage 2 $2SO_2 + O_2 \rightleftharpoons 2SO_3$

Stage 3 production of sulfuric acid from sulfur trioxide

(a) State the meaning of the symbol $\rightleftharpoons$ used in the **stage 2** equation. [1]

....

(b) The temperature used in **stage 2** is 450 °C. This results in a good yield but a low rate.

Name the **compound** used as a catalyst to increase the rate. [1]

....

(c) Dissolving sulfur trioxide in water to form sulfuric acid is too exothermic to be carried out safely in one step.

Give the **two** steps carried out in **stage 3** to convert sulfur trioxide safely into sulfuric acid. [2]

Step 1 ....

....

Step 2 ....

....

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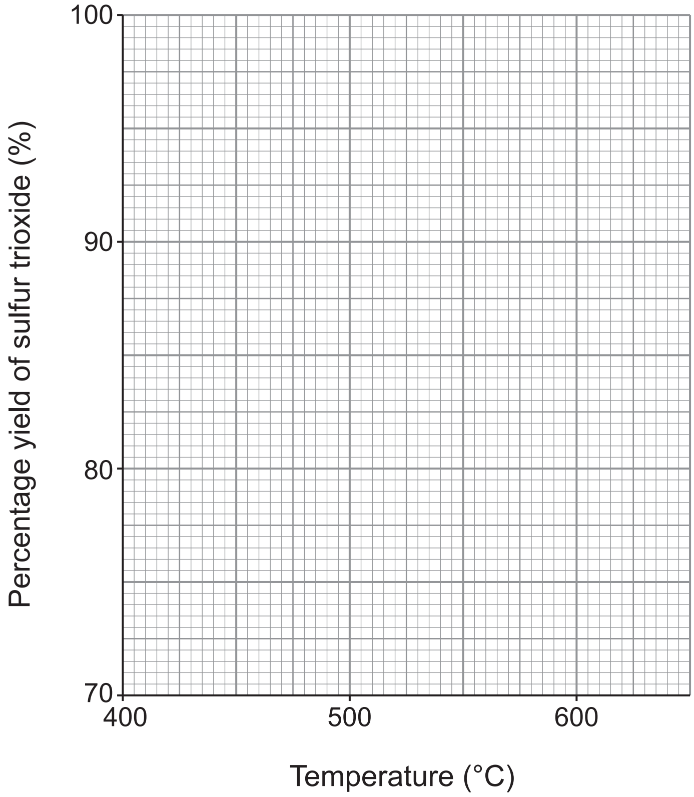
(d) The catalyst used in **stage 2** will **not** work below 400 °C and **breaks down** above 620 °C.

The table shows the yield of sulfur trioxide at different temperatures.

Temperature (°C)	400	450	500	550	600
Percentage yield of sulfur trioxide (%)	99.0	96.5	92.5	86.0	76.5

Plot this data on the grid and draw a suitable line. [3]



(e) State what happens to the percentage yield of sulfur trioxide when the temperature is increased. [1]

....

(f) Use your graph to give the temperature range that produces a yield of 90–99% of sulfur trioxide. [1]

.... to .... °C

9